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Synthesizable HDL Core Specification

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MAC-1G_AMBA

Gigabit Ethernet Media Access Controller

Tests set details

Version 2.02

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1. General informations

All tests are grouped into few categories with respect to their functionality. Although it is possible to change operating frequencies of MAC-1G_AMBA core via generic assignments, all test files are prepared for use with the default settings.

Each test is composed of fourteen files:

/apbstim.txt	: Stimulus file for APB interface
/smemstim.txt	: Stimulus file for shared memory model
/arbstim.txt	: Stimulus file for AMBA™ arbiter model
/linkstim.txt	: Stimulus file for GMII/MII interface
/apbcomp.txt	: Expected Test Bench results from APB interface.
/ahbcomp.txt	: Expected Test Bench results from AHB interface.
/linkcomp.txt	: Expected Test Bench results from GMII/MII.
/intcomp.txt	: Expected Test Bench results from interrupt line.
/apbdiff.txt	: APB interface difference file
/ahbdiff.txt	: AHB interface difference file
/linkdiff.txt	: GMII/MII interface difference file
/intdiff.txt	: Interrupt line difference file.
/time.txt	: Proposed simulation time.
/generic.txt	: Set of generic parameters

For detailed information about testbench configuration refer to MAC-1G_AMBA Media Access Controller Specification.

2. Test sets

The complete MAC-1G_AMBA test suite is divided into several test sets. Each set is chosen to verify the operability of the core in different system and/or network conditions.

The test sets are defined in table 1. Each set consists of 1 or more test groups. The tests groups are defined in the following chapters.

Table 1. MAC-1G_AMBA test sets

test set name	data interface width	MII/GMII interface speed	test groups															
			bd	dma	raf	rfifo	tfifo	rlsm	tlsm	gpt, int	rc, tc	tim	misc	fcr, fct	fcstat	esc	gsc	
d08	8 bit	10/100	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓						
d08_fc	8 bit	10/100											✓	✓				
d08_g	8 bit	1G	✓	✓														
d08_sc	8 bit	10/100													✓			
d16	16 bit	10/100	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓						
d16_fc	16 bit	10/100											✓	✓				
d16_g	16 bit	1G	✓	✓														
d16_sc	16 bit	10/100													✓			
d16_gsc	16 bit	1G														✓		
d32	32 bit	10/100	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓						
d32_fc	32 bit	10/100											✓	✓				
d32_g	32 bit	1G	✓	✓	✓	✓	✓		✓									
d32_sc	32 bit	10/100													✓			
d32_gsc	32 bit	1G														✓		

3. Backoff/Deferring tests

Tests in this group are prepared to test compatibility with IEEE 802.3 standard for Carrier Sense Multiple Access with Collision Detection (CSMA/CD).

Table 2. Backoff/Deferring tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	1 collision	bd1	
2	2 collisions	bd2	
3	3 collisions	bd3	
4	15 collisions	bd4	
5	16 collisions	bd5	
6	late collision	bd6	
7	normal and late collisions	bd7	
8	collision during preamble	bd8	
9	collision during sfd	bd9	
10	collision during crc	bd10	
11	deferring	bd11	

4. Transmit FIFO tests

Tests in this group are prepared to test transmit FIFO operation.

Table 3. Transmit tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	long frame transmission	tfifo1	frame 1 : size = 8192 frame 2 : size = 8191 frame 3 : size = 8194
2	long frame transmission	tfifo2	frame 1 : size = 2047 frame 2 : size = 2047 frame 3 : size = 2047 frame 4 : size = 2047
3	long frame transmission	tfifo3	frame 1 : size = 8192 frame 2 : size = 8192 frame 3 : size = 8192 frame 4 : size = 8192 frame 5 : size = 8192
4	treshold levels: TR=1, TTM=0	tfifo4	frame 1 : size = 2047 frame 2 : size = 2047 frame 3 : size = 2047 frame 4 : size = 2047
5	treshold levels: TR=2, TTM=0	tfifo5	as above
6	treshold levels: TR=3, TTM=0	tfifo6	as above
7	treshold levels: TR=0, TTM=1	tfifo7	as above
8	treshold levels: TR=1, TTM=1	tfifo8	as above
9	treshold levels: TR=2, TTM=1	tfifo9	as above
10	treshold levels: TR=3, TTM=1	tfifo10	as above
11	treshold levels: TR=0, TTM=0, SF=1	tfifo11	as above
12	underflow condition	tfifo12	frame 1 : size = 2047und. frame 2 : size = 2047 frame 3 : size = 2047und. frame 4 : size = 2047und. frame 5 : size = 2047

5. Transmit linked list state machine tests

Tests in this group are prepared to test linked list state machine which is part of MACs Descriptor/Buffer architecture.

Table 4. Transmit linked list state machine tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	1 buffer, various frame sizes for address(2..0)=000, for buf1	tlsm1	
2	1 buffer, various address alignments for size(2..0)=000, for buf1	tlsm2	
3	1 buffer, various frame sizes for address(2..0)=000, for buf2	tlsm3	
4	1 buffer, various address alignments for size(2..0)=000, for buf2	tlsm4	
5	1 buffer, various frame sizes for address(2..0)=001, for buf1	tlsm5	
6	1 buffer, various frame sizes for address(2..0)=010, for buf1	tlsm6	
7	1 buffer, various frame sizes for address(2..0)=011, for buf1	tlsm7	
8	1 buffer, various frame sizes for address(2..0)=100, for buf1	tlsm8	
9	1 buffer, various frame sizes for address(2..0)=101, for buf1	tlsm9	
10	1 buffer, various frame sizes for address(2..0)=110, for buf1	tlsm10	
11	1 buffer, various frame sizes for address(2..0)=111, for buf1	tlsm11	
12	1 buffer, various frame sizes for address(2..0)=001, for buf2	tlsm12	
13	1 buffer, various frame sizes for address(2..0)=010, for buf2	tlsm13	
14	1 1 buffer, various frame sizes for address(2..0)=011, for buf2	tlsm14	
15	1 buffer, various frame sizes for address(2..0)=100, for buf2	tlsm15	
16	1 buffer, various frame sizes for address(2..0)=101, for buf2	tlsm16	
17	1 buffer, various frame sizes for address(2..0)=110, for buf2	tlsm17	
18	1 buffer, various frame sizes for address(2..0)=111, for buf2	tlsm18	
19	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=000"	tlsm19	
20	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=001"	tlsm20	
21	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=010"	tlsm21	
22	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=011"	tlsm22	
23	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=100"	tlsm23	
24	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=101"	tlsm24	
25	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=110"	tlsm25	
26	2 buffers, various buffer 1 sizes for buffer 2 address(2..)=111"	tlsm26	
27	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=000"	tlsm27	
28	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=001"	tlsm28	
29	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=010"	tlsm29	
30	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=011"	tlsm30	
31	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=100"	tlsm31	
32	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=101"	tlsm32	
33	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=110"	tlsm33	
34	2 descriptors/2 buffers, various buffer 1 sizes for buffer 2 address(2..)=111"	tlsm34	
35	2 descriptors/3 buffers per frame	tlsm35	
36	2 descriptors/4 buffers per frame	tlsm36	
37	3 descriptors per frame	tlsm37	
38	7 descriptors per frame	tlsm38	
39	7 descriptors per frame, chain mode	tlsm39	
40	empty descriptors	tlsm40	

6. Transmit interrupt mitigation control tests

Tests in this group are prepared to test transmit interrupt mitigation control.

Table 5. Transmit interrupt mitigation control tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	TT=1, NRP=0, CC=1	tim1	
2	TT=1, NRP=7, CC=1	tim2	
3	TT=0, NRP=7, CC=1	tim3	
4	TT=1, NRP=7, CC=1, IC=1	tim4	
5	TT=2, NRP=0, CC=1	tim5	
6	TT=3, NRP=0, CC=1	tim6	
7	TT=14, NRP=0, CC=1	tim7	
8	TT=15, NRP=0, CC=1	tim8	
9	TT=2, NRP=0, CC=0	tim9	
10	TT=3, NRP=0, CC=0	tim10	
11	TT=0, NRP=1, CC=1	tim11	
12	TT=0, NRP=2, CC=1	tim12	
13	TT=0, NRP=3, CC=1	tim13	
14	TT=0, NRP=4, CC=1	tim14	

7. Receive operation tests

Tests in this group are prepared to test receive operation.

Table 6. Receive operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	ethernet frame with 1517 bytes	rc1	
2	ethernet frame with 1519 bytes	rc2	
3	802.3 frame with 1517 bytes	rc3	
4	802.3 frame with 1519 bytes	rc4	
5	frame with 8193 bytes	rc5	
6	pass bad frame mode - frame with 63 bytes	rc6	
7	frame with crc error	rc7	
8	frame with data alignment error	rc 8	
9	frame with mii error	rc 9	
10	receiving in the suspended/stopped state	rc 10	

8. Receive address filtering tests

Tests in this group are prepared to test receive address filtering.

Table 7. Receive address filtering tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	perfect filtering	raf1	
2	perfect filtering – filtering fail	raf2	
3	inverse perfect filtering	raf3	
4	hash filtering	raf4	
5	hash only filtering	raf5	
6	receive all mode	raf6	
7	promiscuous mode	raf7	
8	pass all multicast mode	raf8	
9	external address filtering	raf9	

9. Receive FIFO tests

Tests in this group are prepared to test receive FIFO operation.

Table 8. Receive FIFO operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	various frame sizes	rfifo1	frame 1 : size = 68 frame 2 : size = 69 frame 3 : size = 70 frame 4 : size = 71 frame 5 : size = 72 frame 6 : size = 73 frame 7 : size = 74 frame 8 : size = 75
2	various frame sizes	rfifo2	frame 1 : size = 1516 frame 2 : size = 1517 frame 3 : size = 1518 frame 4 : size = 1519 frame 5 : size = 1520 frame 6 : size = 1521 frame 7 : size = 1522 frame 8 : size = 1523
3	various frame sizes	rfifo3	frame 1 : size = 2044 frame 2 : size = 2045 frame 3 : size = 2046 frame 4 : size = 2047 frame 5 : size = 2048 frame 6 : size = 2049 frame 7 : size = 2050 frame 8 : size = 2051
4	various frame sizes	rfifo4	frame 1 : size = 8190 frame 2 : size = 8191 frame 3 : size = 8192 frame 4 : size = 8193 frame 5 : size = 8194 frame 6 : size = 8195 frame 7 : size = 8196 frame 8 : size = 8197
5	various frame sizes	rfifo5	frame 1 : size = 2048 frame 2 : size = 2048 frame 3 : size = 2048 frame 4 : size = 2048 frame 5 : size = 2048 frame 6 : size = 2048 frame 7 : size = 2048 frame 8 : size = 2048
6	overflow condition	rfifo6	frame 1 : size = 8190ove. frame 2 : size = 8190 frame 3 : size = 8190ove. frame 4 : size = 8190ove. frame 5 : size = 8190

10. Receive linked list state machine tests

Tests in this group are prepared to test receive linked list state machine which is part of the MAC's Descriptor/Buffer architecture.

Table 9. Receive operation tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	single buffer, various frame sizes for address(2..0)=111, for buf2	rlsm1	
2	2 buffers per frame	rlsm2	
3	2 descriptors/2 buffers per frame	rlsm3	
4	2 descriptors/3 buffers per frame	rlsm4	
5	2 descriptors/4 buffers per frame	rlsm5	
6	3 descriptors per frame	rlsm6	
7	7 descriptors per frame	rlsm7	
8	7 descriptors per frame, chain mode	rlsm8	
9	empty descriptors	rlsm 9	
10	descriptors unavailable	rlsm10	

11. Receive interrupt mitigation control tests

Tests in this group are prepared to test receive interrupt mitigation control.

Table 10. Receive interrupt mitigation control tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	RT=3, NRP=0, CC=1	rim1	
2	RT=4, NRP=7, CC=1	rim2	
3	RT=0, NRP=7, CC=1	rim3	
4	RT=4, NRP=0, CC=1	rim4	
5	RT=5, NRP=0, CC=1	rim5	
6	RT=14, NRP=0, CC=1	rim6	
7	RT=15, NRP=0, CC=1	rim7	
8	RT=2, NRP=0, CC=0	rim8	
9	RT=3, NRP=0, CC=0	rim9	
10	RT=0, NRP=1, CC=0	rim10	
11	RT=0, NRP=2, CC=0	rim11	
12	RT=0, NRP=3, CC=0	rim12	

12. Interrupt tests

Tests in this group are prepared to test interrupts.

Table 11. Interrupt tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	early receive interrupt	int1	
2	receive interrupt	int2	
3	receive buffer unavailable interrupt	int3	
4	receive process stopped interrupt	int4	
5	early transmit interrupt	int5	
6	transmit interrupt	int6	
7	transmit buffer unavailable interrupt	int7	
8	transmit process stopped interrupt	int8	

13. General purpose timer tests

Tests in this group are prepared to test general purpose timer.

Table 12. General purpose timer tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	TIM=65535, CON=0	gpt1	
2	TIM=1024, CON=0	gpt 2	
3	TIM=1023, CON=0	gpt 3	
4	TIM=1022, CON=0	gpt 4	
5	TIM=3, CON=0	gpt 5	
6	TIM=2, CON=0	gpt 6	
7	TIM=1, CON=0	gpt 7	
8	TIM=65535, CON=1	gpt 8	
9	TIM=1024, CON=1	gpt 9	
10	TIM=1023, CON=1	gpt 10	
11	TIM=1022, CON=1	gpt 11	
12	TIM=3, CON=1	gpt 12	
13	TIM=2, CON=1	gpt 13	
14	TIM=1, CON=1	gpt 14	

14. DMA tests

Tests in this group are prepared to test Direct Memory Access operation.

Table 13. DMA tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	burst length = 0, arbitration mode 1	dma1	
2	burst length = 1, arbitration mode 1	dma2	
3	burst length = 2, arbitration mode 1	dma3	
4	burst length = 4, arbitration mode 1	dma4	
5	burst length = 8, arbitration mode 1	dma5	
6	burst length = 16, arbitration mode 1	dma6	
7	burst length = 32, arbitration mode 1	dma7	
8	burst length = 0, arbitration mode 2	dma8	
9	burst length = 1, arbitration mode 2	dma9	
10	burst length = 2, arbitration mode 2	dma10	
11	burst length = 4, arbitration mode 2	dma11	
12	burst length = 8, arbitration mode 2	dma12	
13	burst length = 16, arbitration mode 2	dma13	
14	burst length = 32, arbitration mode 2	dma14	
15	big endian mode for descriptors	dma15	
16	big endian mode for buffers	dma16	
17	big endian mode for descriptors/buffers	dma17	
18	wait cycles for burst length = 0	dma18	
19	wait cycles for burst length = 1	dma19	

15. Miscellaneous tests

Tests in this group are prepared to test functionality not covered by functional tests.

Table 14. Miscellaneous tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	transmission with $\text{clkdma}, \text{clkcsr} = \text{clkmi}/10$	misc1	
2	transmission with $\text{clkdma}, \text{clkcsr} = 10 \cdot \text{clkmi}$	misc2	
3	transmission with $\text{clkdma} = \text{clkmi} = 10 \cdot \text{clkcsr}$	misc3	
4	transmission with $\text{clkdma} = \text{clkmi} = \text{clkcsr}/10$	misc4	
5	receiving with $\text{clkdma}, \text{clkcsr} = \text{clkmi}/10$	misc5	
6	receiving with $\text{clkdma}, \text{clkcsr} = 10 \cdot \text{clkmi}$	misc6	
7	receiving with $\text{clkdma} = \text{clkmi} = 10 \cdot \text{clkcsr}$	misc7	
8	receiving with $\text{clkdma} = \text{clkmi} = \text{clkcsr}/10$	misc8	
9	simultaneous transmission/receiving with $\text{clkdma}, \text{clkcsr} = \text{clkmi}/10$	misc9	
10	simultaneous transmission/receiving with $\text{clkdma}, \text{clkcsr} = 10 \cdot \text{clkmi}$	misc10	
11	simultaneous transmission/receiving with $\text{clkdma} = \text{clkmi} = 10 \cdot \text{clkcsr}$	misc11	
12	simultaneous transmission/receiving with $\text{clkdma} = \text{clkmi} = \text{clkcsr}/10$	misc12	

16. Flow control receive tests

Tests in this group are prepared to test functionality of flow control receiver.

Table 15. Flow control receive tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	FC receiver setup and status test	fcr1	
2	FC transmit delaying on request	fcr2	
3	FC transmit delaying cancellation on request	fcr3	
4	FC receiving frames with not globally assigned address	fcr4	
5	FC pause condition during transmission process handling	fcr5	

17. Flow control transmit tests

Tests in this group are prepared to test functionality of flow control transmitter.

Table 16. Flow control transmit tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	FC transmitter setup -HD/FD TUE TPE handling	fct1	
2	FC transmitter setup - FCE handling	fct2	
3	FC transmitting pause cache thresholds setup	fct3	
4	FC transmitting pause frame thresholds setup	fct4	
5	FC transmitter pause frame insertion in output stream	fct5	

18. Flow control statistical module tests

Tests in this group are prepared to test functionality of flow control statistical module.

Table 17. Flow control statistical module tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	FC transmit statistical counters test	fcstat1	
2	FC receive statistical counters test	fcstat2	

19. Common statistical counters tests

Tests in this group are prepared to test functionality of common statistical counters module.

Table 18. Common statistical tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	statistical counters test - transmit	esc1	
2	statistical counters test - receive	esc2	
3	errors handling of the transmit frame for half duplex mode	esc3	
4	errors handling of the transmit frame for full duplex mode	esc4	
5	excessive collision test	esc5	
6	receive frame type detection	esc6	
7	receiving frames of various sizes	esc7	
8	errors handling of the receive frame	esc8	
9	transmit statistical counters clearing	esc9	
10	receive statistical counters clearing	esc10	
11	statistical counters clearing - various situations	esc11	

20. Gigabit mode statistical counters tests

Tests in this group are prepared to test functionality of statistical counters module with gigabit mode counters.

Table 19. Gigabit mode statistical tests summary

No.	Purpose/conditions	Test name	Limitations/Comments
1	gigabit mode statistical counter test - transmit	gsc1	
2	gigabit mode statistical counter test - receive	gsc2	